Design and Promotion of Medicinal Diet Menus for Myopia Prevention in Jiangsu School Cafeterias

——*Could My Cafeteria Tray Improve My Vision?*

Team Name: EaglePlate Team
Team Captain: Tang Ziyi



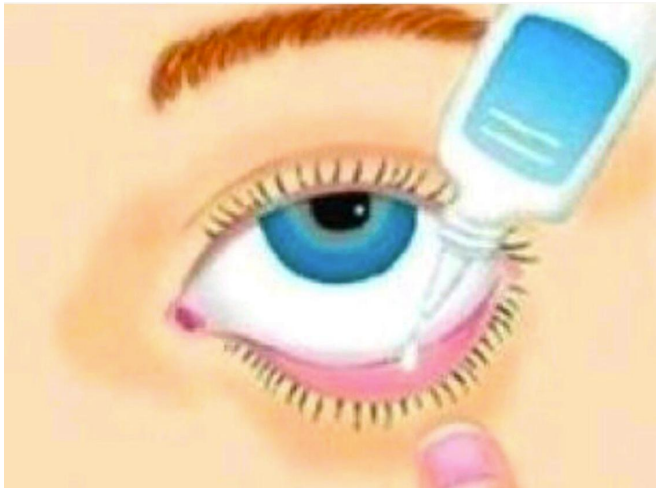
1 Current Strategies

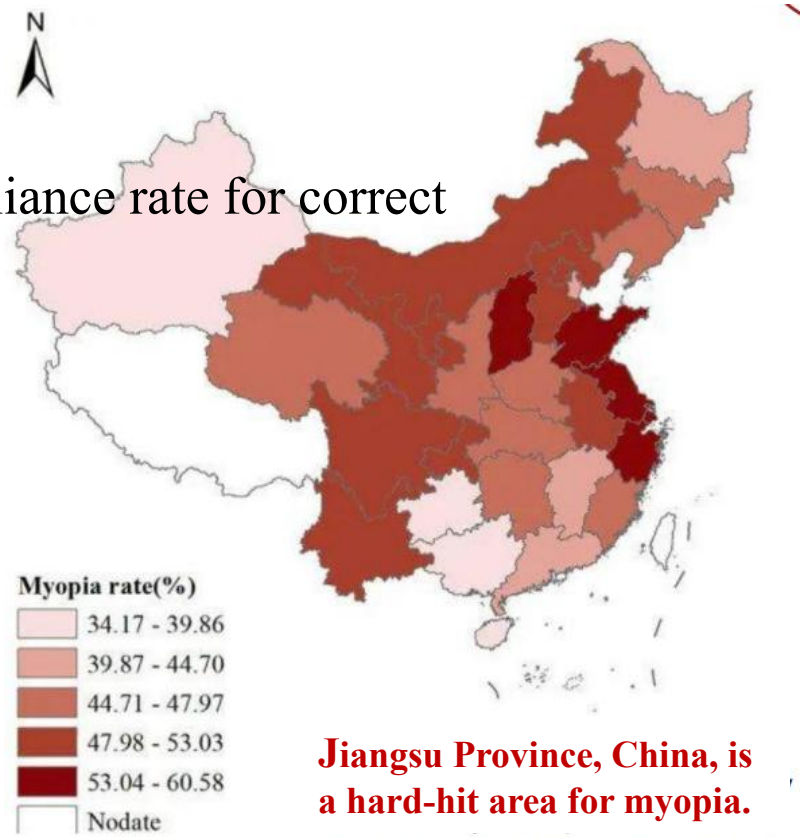


1 Current Strategies

- Academic pressures leave little time for outdoor activities.
- Atropine eye drops can cause pupil dilation.
- Orthokeratology lenses are expensive and carry a risk of keratitis.
- In eye exercises, acupoint location accuracy is just 50.7%, and the compliance rate for correct pressure application is a mere 30.6%.





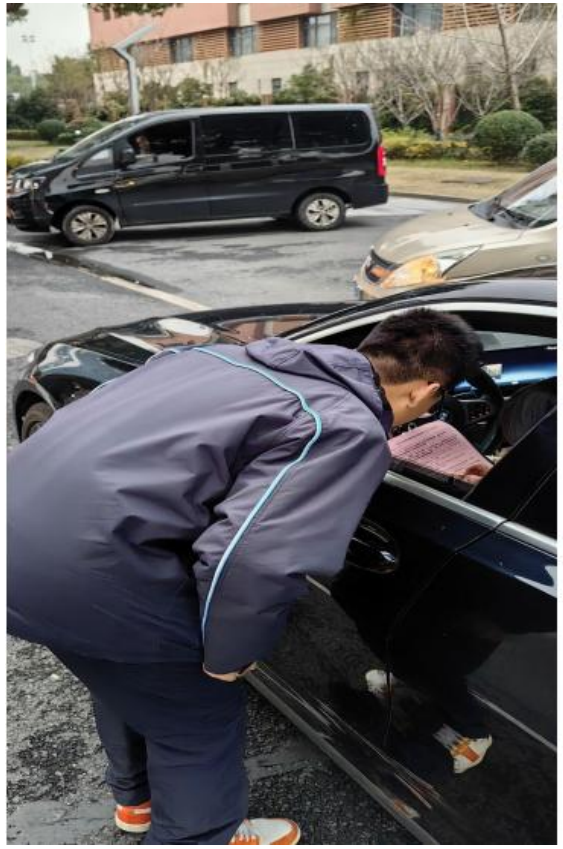


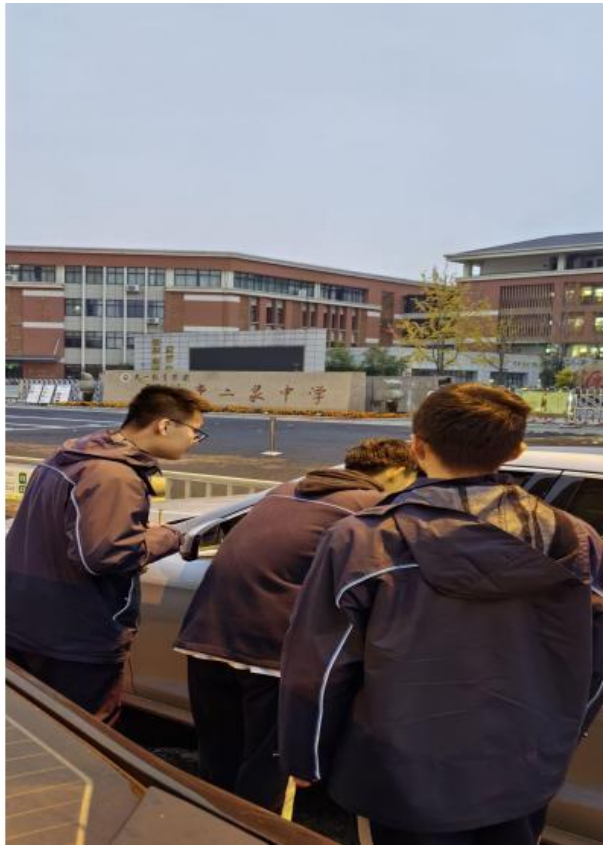
Jiangsu Province, China, is a hard-hit area for myopia.

[1] Cai, R. M. (2024). A school-based study on the causes of and countermeasures for myopia in primary school students. Journal of the Chinese Society of Education, S2, 1–2.
[2] Chen, X. J., Zhao, W. X., Zhang, C. H., et al. (2024). A cross-sectional survey on the correct practice of eye exercises among school-age children. Chinese Journal of School Doctor, 38(8), 577–580.



2 Preliminary Research





2.1 Literature Review

（1）Traditional Chinese Medicine Advocates the Principle of "Medicine and Food Sharing the Same Origin, Herbs and Food Sharing the Same Roots"

- China's dietary therapy for eye diseases can be traced back to ancient times.
- Over a 3,000-year history of dietary therapy, our ancestors accumulated rich experience in preventing and treating eye diseases through diet.
- The concept of food-medicine homology provides an important reference for modern myopia prevention and control.

Period	Overview of Dietary Therapy Development for Eye Diseases	Key Literature and Works
Antiquity to the Northern and Southern Dynasties	Embryonic stage	Huangdi Neijing (Inner Canon of the Yellow Emperor), Shennong Bencao Jing (Shennong's Classic of Materia Medica), Shanghan Zabing Lun (Treatise on Cold Damage and Miscellaneous Diseases)
Sui and Tang Dynasties	Increase in ophthalmology texts	Qianjin Yaofang (Essential Prescriptions Worth a Thousand Gold), et al.
Song and Yuan Dynasties	Ophthalmology became an independent specialty	Taiping Shenghui Fang (Taiping Holy Prescriptions for Universal Relief), Shengji Zonglu (Comprehensive Recording of Divine Assistance), et al.
Ming and Qing Dynasties to the Eve of the Opium War	Reached its peak	Yicaoting Muke Quanshu (Complete Book of Ophthalmology of Yicaoting), Shenshi Yaohan (Collected Works of Ophthalmology), et al.
Modern Era	Decline stage	Yanke Jinjing (Golden Mirror of Ophthalmology), Yanke Jinghua Lu (Essence of Ophthalmology), et al.
After the Founding of New China	Integration of traditional Chinese and Western medicine	Encyclopedia of Chinese Medicine: Traditional Chinese Medicine Ophthalmology, et al.



The overall level of traditional Chinese medicine in Jiangsu ranks among the top tier nationwide.

2.1 Literature Review

（2）Modern Medical Science Demonstrates a Close Link Between Diet and Myopia

- Gardiner (1960) first proposed that diet can improve myopia.
- Masenya (2022) found that diet is the second largest factor affecting myopia, just behind outdoor activity.
- Yin (2023) confirmed that the Jiangnan dietary pattern can lower the risk of myopia.


Vegetables


Fruits


Fish


Seafood


Legumes


Nuts

[3] Yin, C. J. (2022). Nutrients and myopia. Journal of Hygiene Research, 51(5), 720–724.
[4] Gardiner, P. A. (1960). Protein and myopia. Proceedings of the Nutrition Society, 19(1), 96-100.
[5] Masenya, A. O. (2022). Role Of Nutrition on Myopia Progression. Thesis submitted to the Department of Optometry, School of Advanced Education, Research and Accreditation (SAERA), Spain. Supervised by Raúl Pérez.
[6] Yin, C., Gan, Q., Xu, P., Yang, T., Xu, J., Cao, W., ... & Zhang, Q. (2023). Dietary patterns and associations with myopia in Chinese children. Nutrients, 15(8), 1946

2.1 Literature Review

（3）Official Documents from the National Health Commission and Other Authorities

- Ages 0–17 are a critical period for visual development.
- It recommends eating more fish, fresh fruits, vegetables, etc.



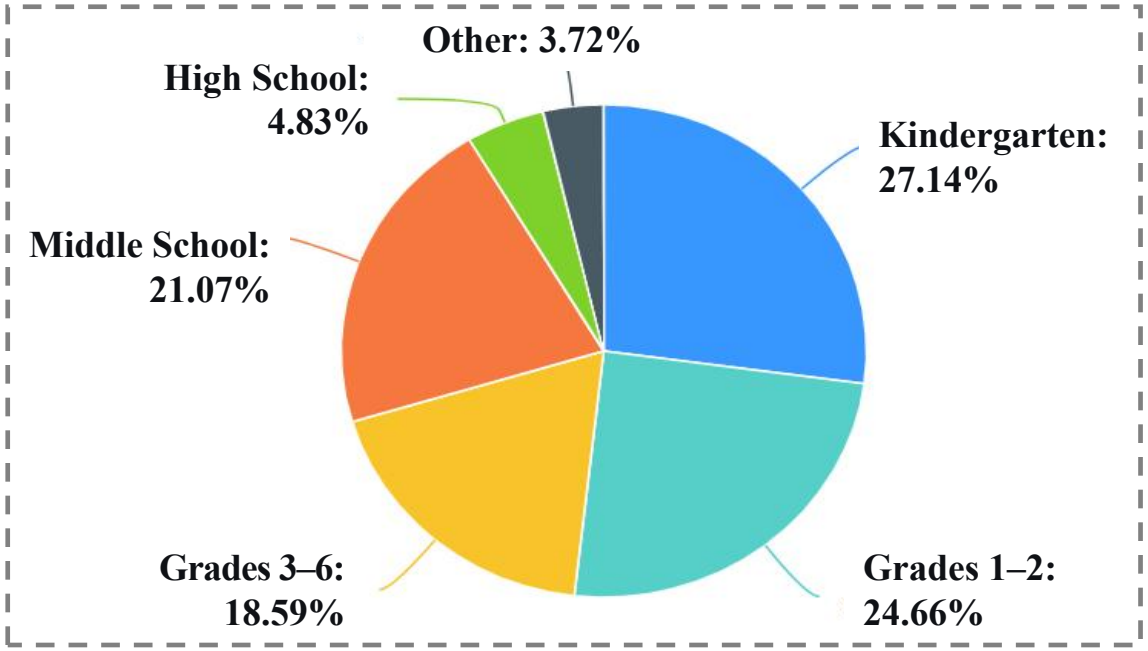
[7] General Office of the National Health Commission. (2023). Notice on issuing the ten core knowledge points for the prevention and control of myopia in children and adolescents. Retrieved from the National Health Commission website: http://www.nhc.gov.cn/fys/s7906/202307/7a37c015f42340cf9387be594662bb73.shtml

[8] Ministry of Education and Seven Other Departments. (2018). Notice on issuing the Implementation Plan for Comprehensive Prevention and Control of Myopia in Children and Adolescents [Policy document]. Retrieved from: http://www.moe.gov.cn/srcsite/A17/moe_943/s3285/201808/t20180830_346672.html

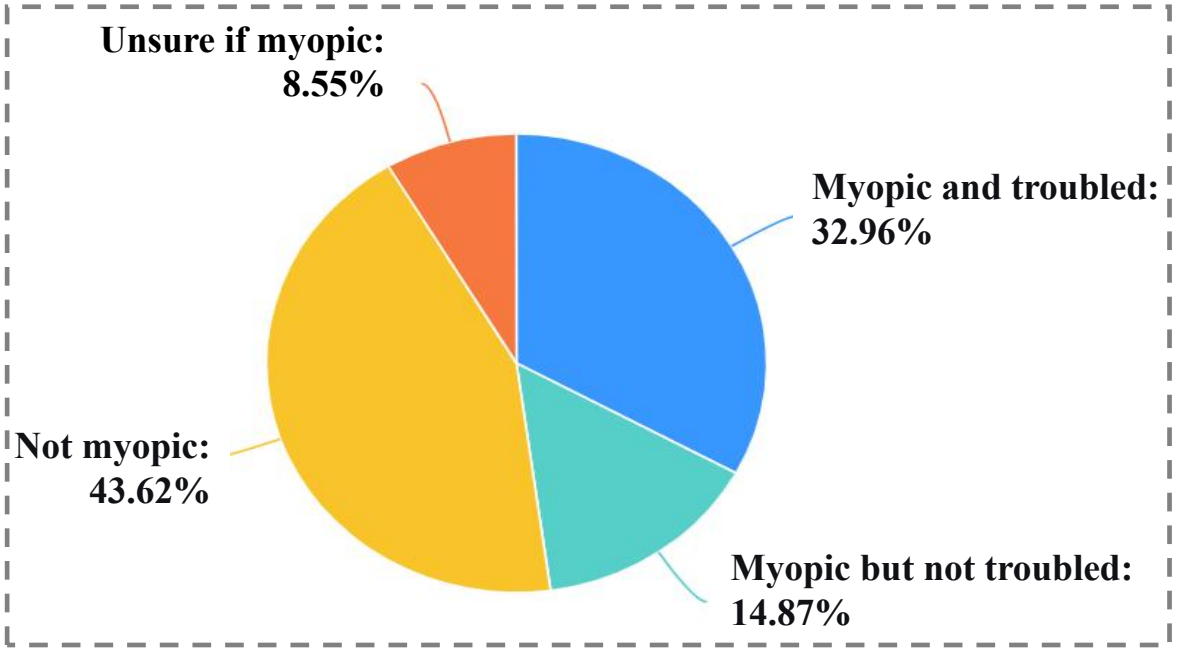
2.2 Questionnaire Survey

（1）Myopia Status of the Survey Sample

- The respondents were mostly kindergarten and lower elementary school students, with a total of 807 participants.
- The survey revealed that myopia is a widespread issue among students: only 43.62% reported having normal vision.



Students' Grade Levels

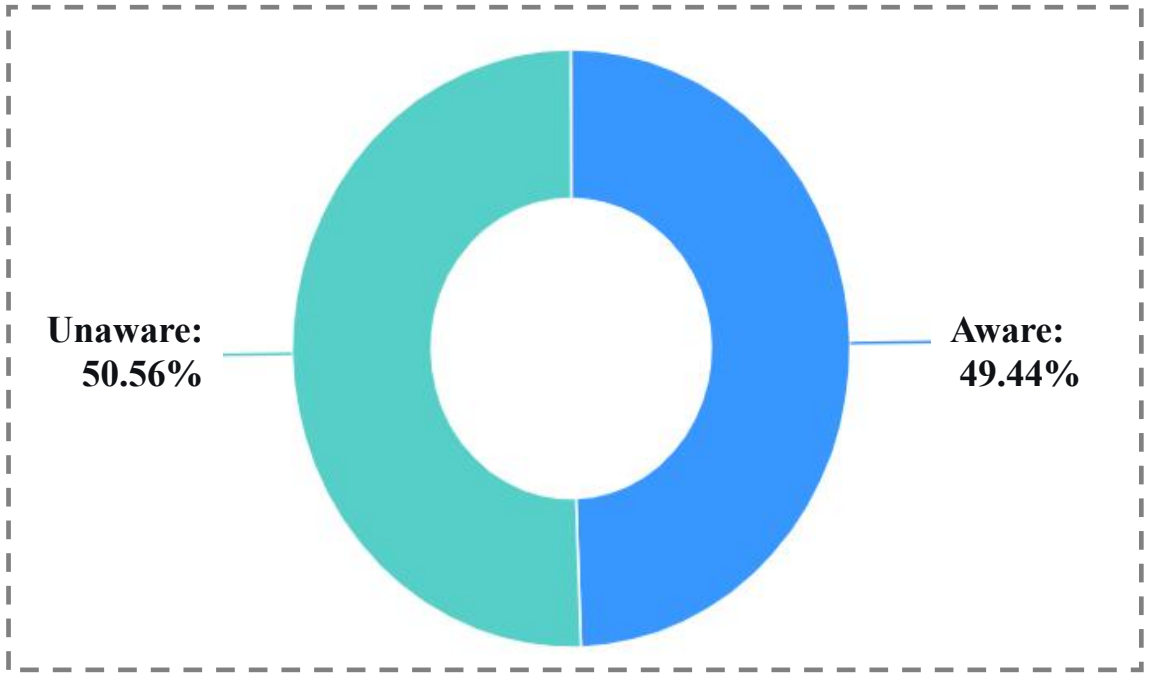


Students' Myopia Status and Concerns

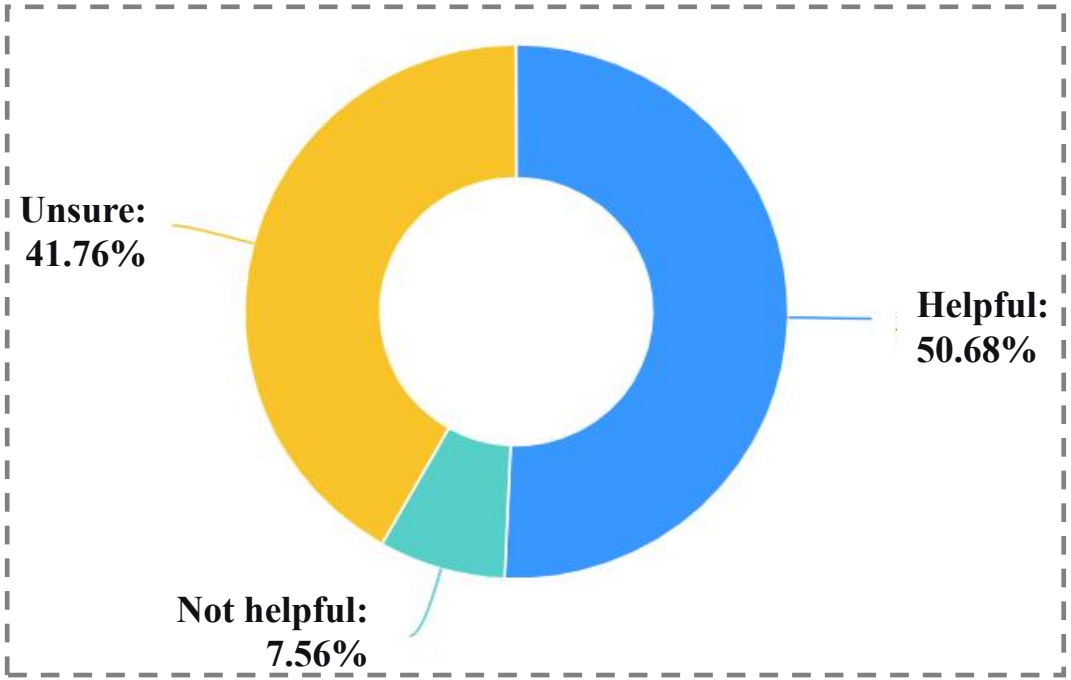
2.2 Questionnaire Survey

（2）Awareness of Food-Medicine Homology and Myopia Prevention

- Although the questionnaire explained the concept of "food-medicine homology," only 50.56% of respondents understood it, and just 50.68% believed it was helpful for preventing myopia.



Awareness of the "food-medicine homology" concept

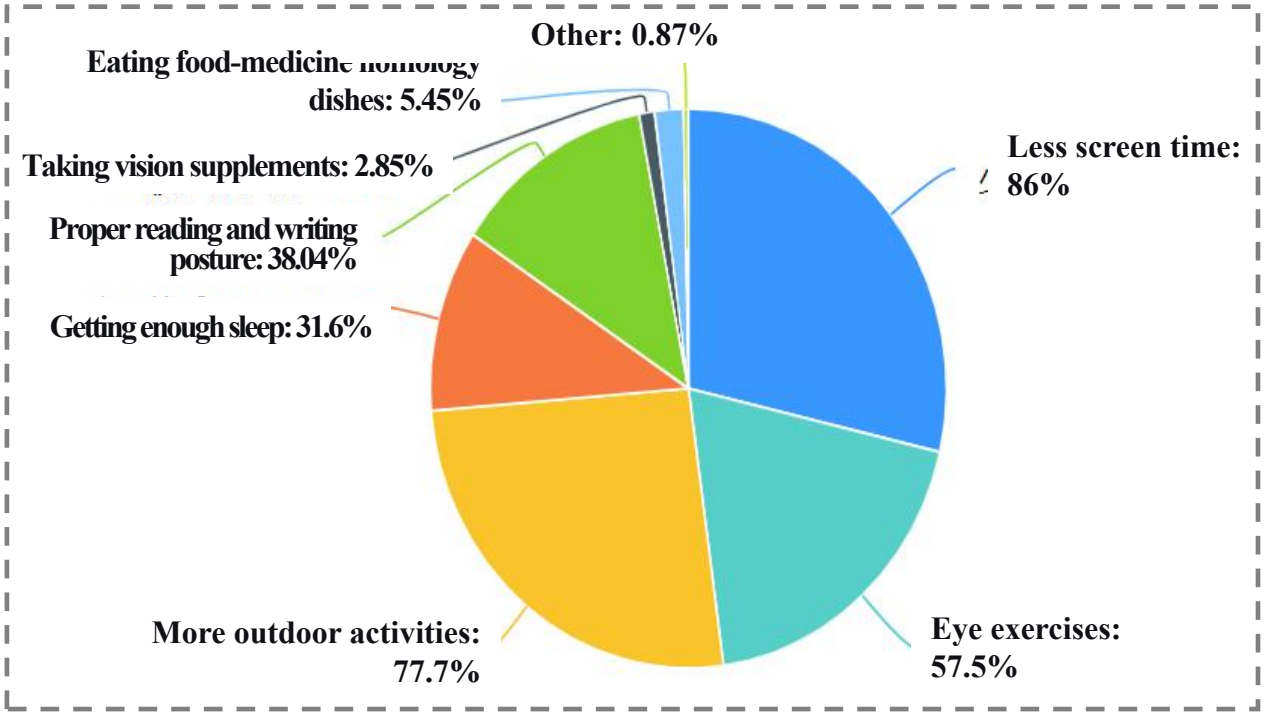


Perceived Effectiveness of Dietary Therapy in Preventing Myopia

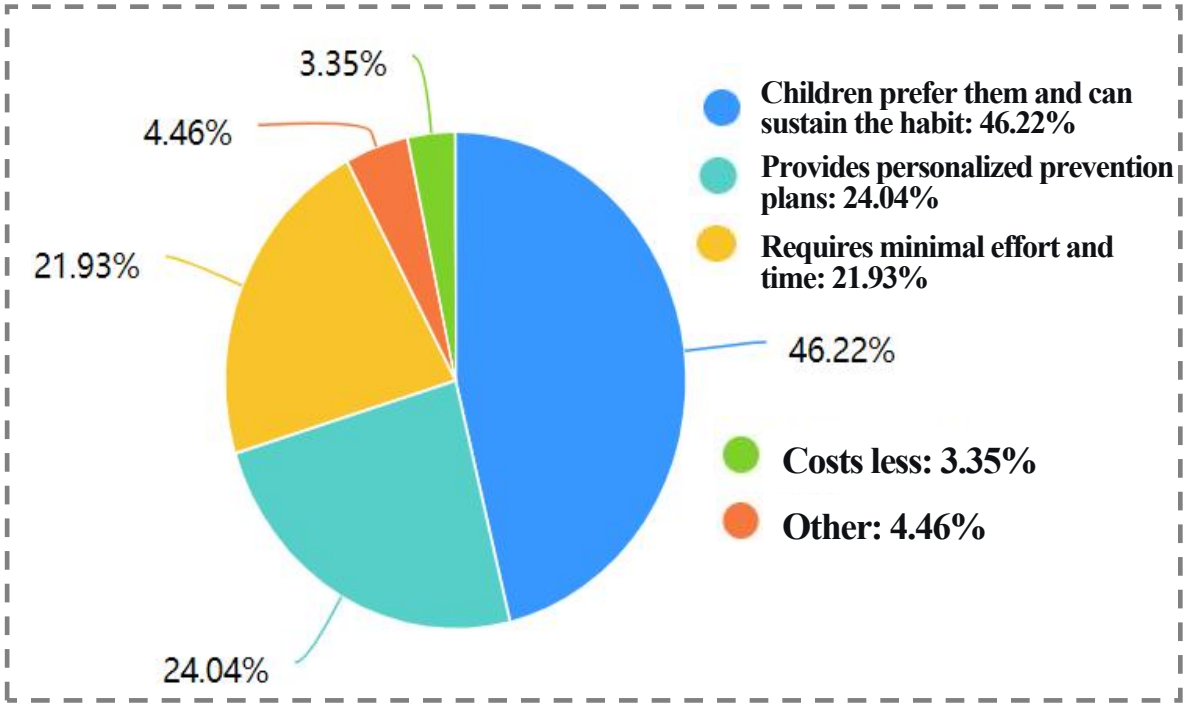
2.2 Questionnaire Survey

（2）Awareness of Food-Medicine Homology and Myopia Prevention

• Among myopia prevention methods, limiting screen time, increasing outdoor activities, and doing eye exercises are widely recognized. However, awareness of food-medicine homology dishes remains low at just 5.45%. Still, respondents believe these dishes offer potential advantages, particularly in terms of children's acceptance and personalized prevention.



Acceptance of Myopia Prevention Methods

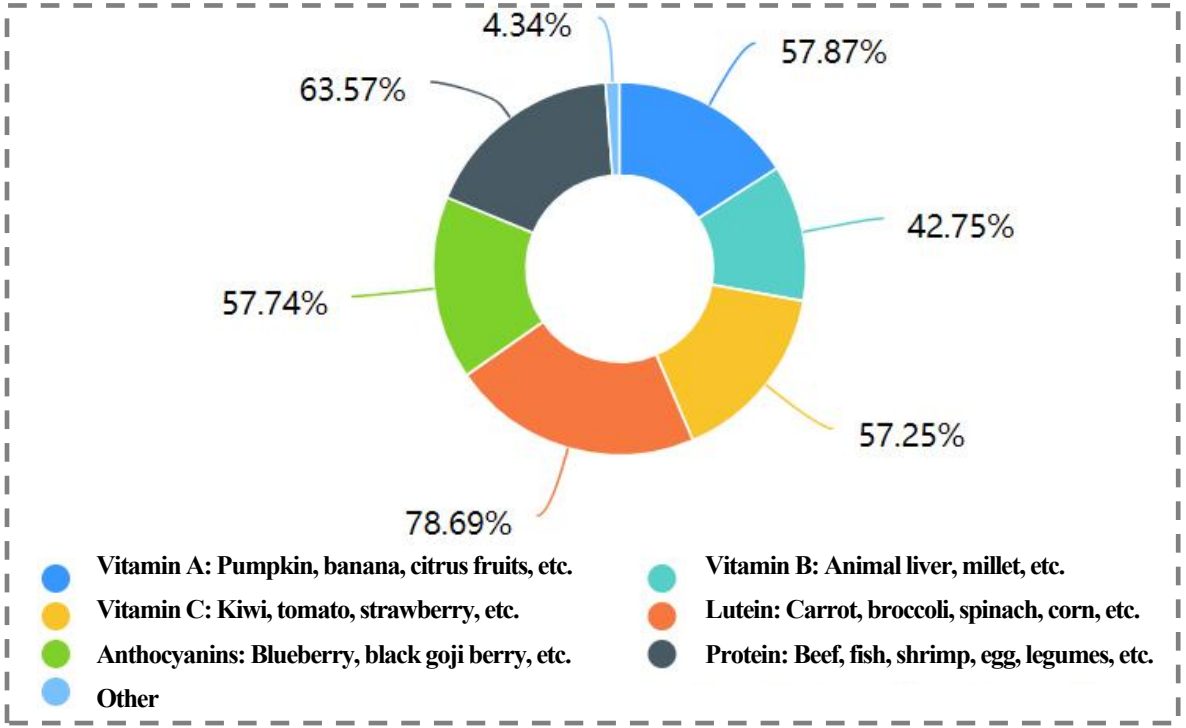


Advantages of Food-Medicine Homology Dishes for Myopia Prevention

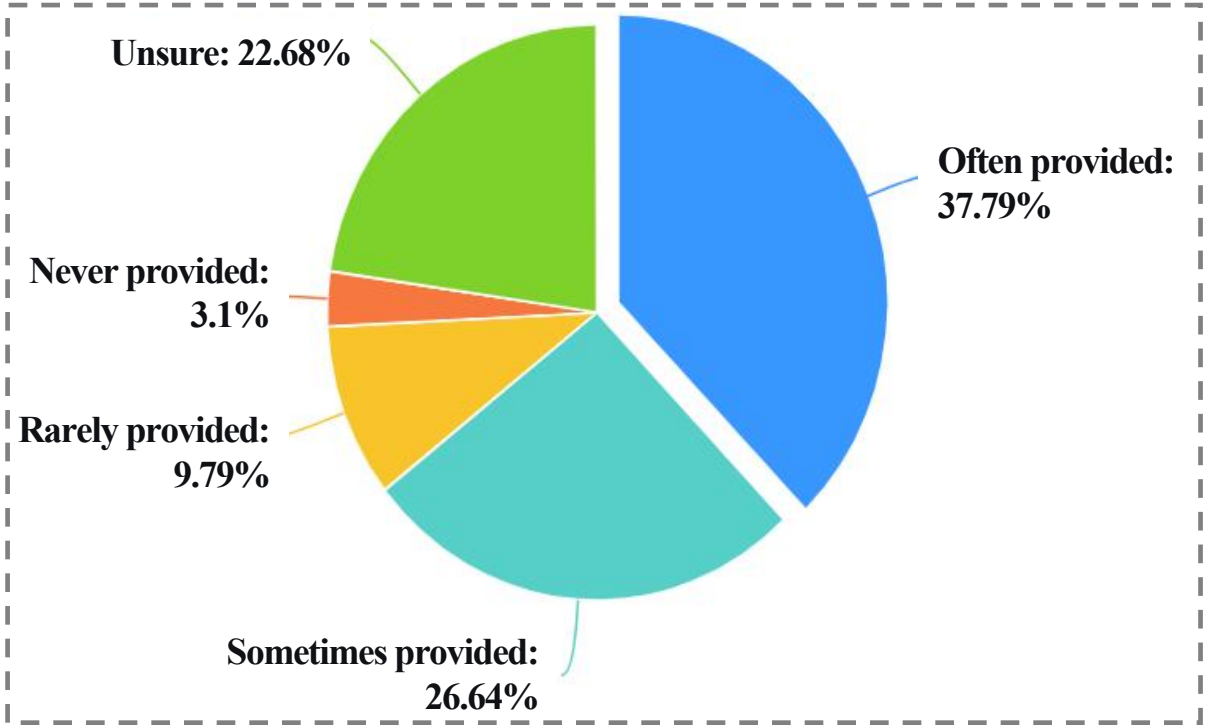
2.2 Questionnaire Survey

（3）Current Status of Myopia-Preventing Dish Provision in School Cafeterias

- School cafeterias fall short in supplying key nutrients essential for myopia prevention, such as lutein, protein, and vitamins A and C. Only 37.79% of respondents reported that their schools frequently provide foods with myopia-preventing benefits.



Ingredients Students Get the Least of in the School Cafeteria

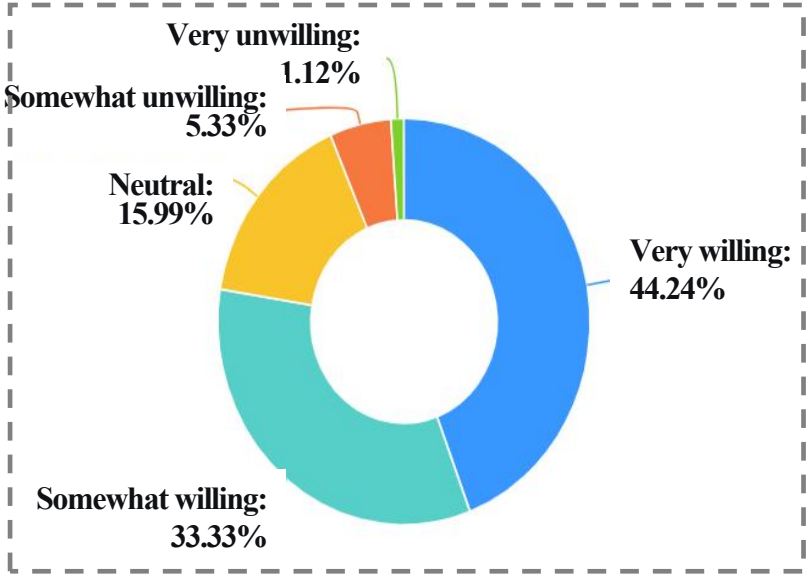


Frequency of Myopia-Preventing Dishes Provided in School Cafeterias

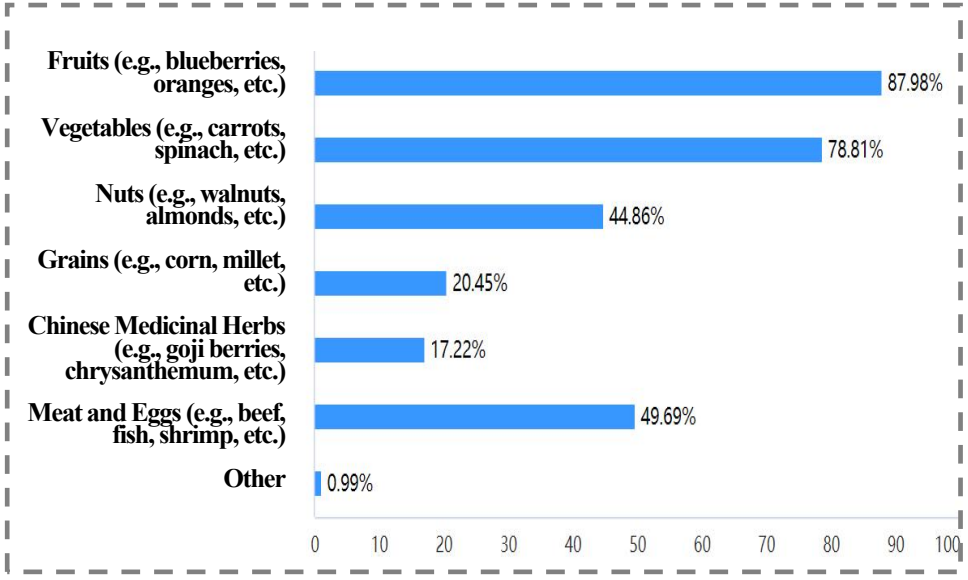
2.2 Questionnaire Survey

（4）Purchase Intent for Food-Medicine Homology Dishes

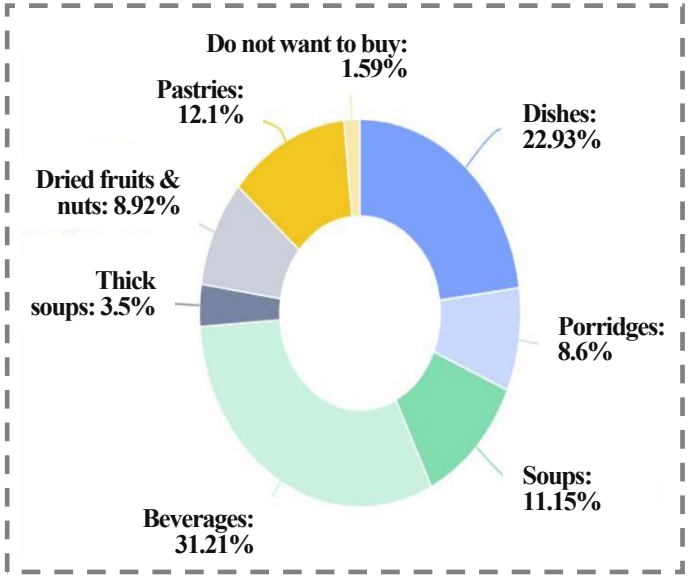
- Respondents showed a strong willingness to purchase food-medicine homology dishes and clearly preferred natural ingredients like **fruits**, **vegetables**, and **nuts**. The most popular type of dish was **beverages**.



Purchase Intent for Myopia-Preventing Food-Medicine Homology Dishes



Preferred Ingredients for Food-Medicine Homology Dishes

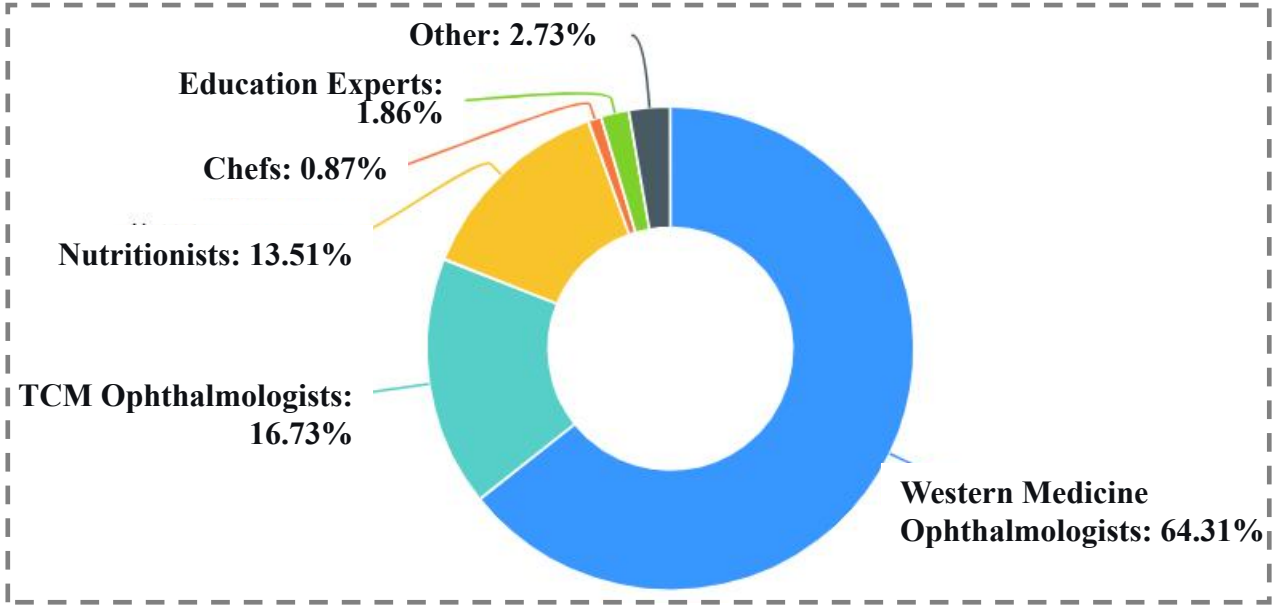


Most Preferred Types of Food-Medicine Homology Dishes

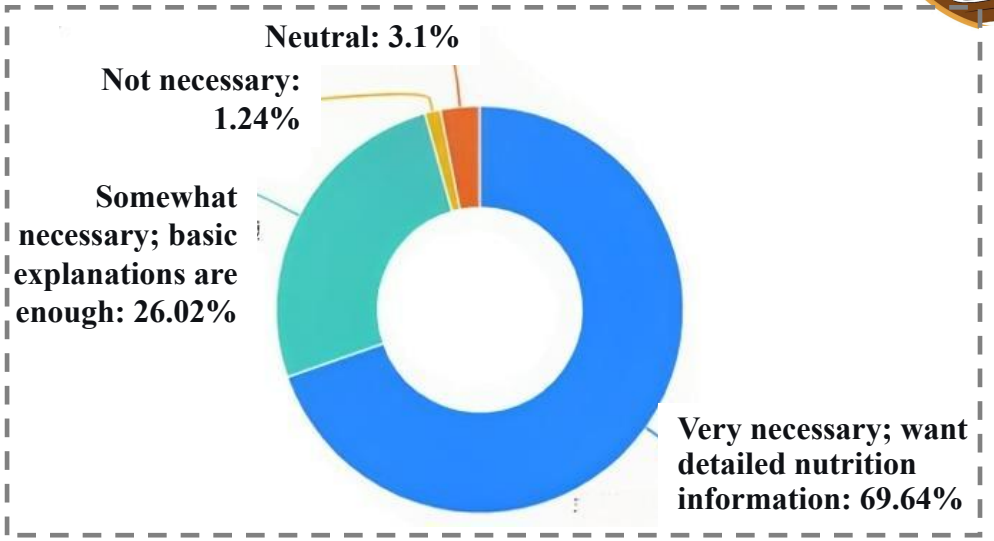
2.2 Questionnaire Survey

（5）Expectations Regarding Food-Medicine Homology Recipe Knowledge

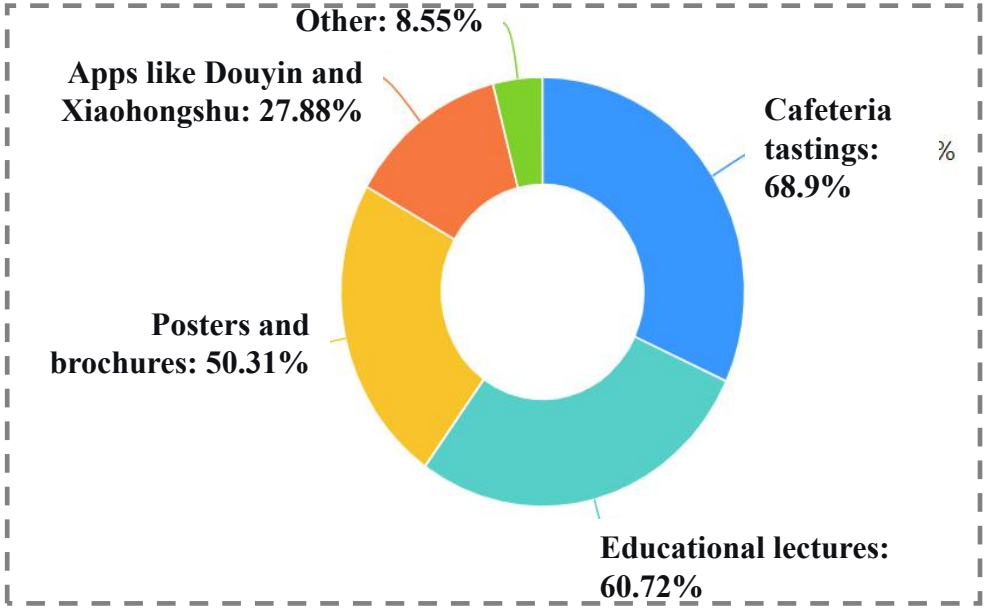
• Respondents showed a strong demand for nutritional knowledge, but when it came to seeking recipe recommendations, their trust in traditional Chinese medicine (TCM) was noticeably lower than in Western medicine. They preferred to learn about food-medicine homology dishes through cafeteria tastings, educational lectures, and posters.



Preferred Experts for Food-Medicine Homology Recipe Recommendations



Need for Nutritional Knowledge on Myopia-Preventing Dishes



Ways to Learn About Myopia-Preventing Dishes in School Cafeterias



3 Innovation Actions



3.1 Recipe Compilation

Multiple rounds of screening for myopia-preventing food-medicine homology recipes

- Reviewed 53 books.
- Compiled "83 Food-Medicine Homology Recipes for Myopia Prevention."
- Compiled "55 Questions on Dietary Therapy Knowledge for Myopia Prevention."
- The safety of the recipes was reviewed by experts in the field and the food ingredient company.



Dr. Huang Xiuxian, Department of Preventive Treatment of Disease, Zhenjiang Hospital of Integrated Traditional Chinese and Western Medicine



Dr. Sun Dewei, Vice Dean of the School of Tourism and Culinary Arts, Wuxi Vocational Institute of Commerce

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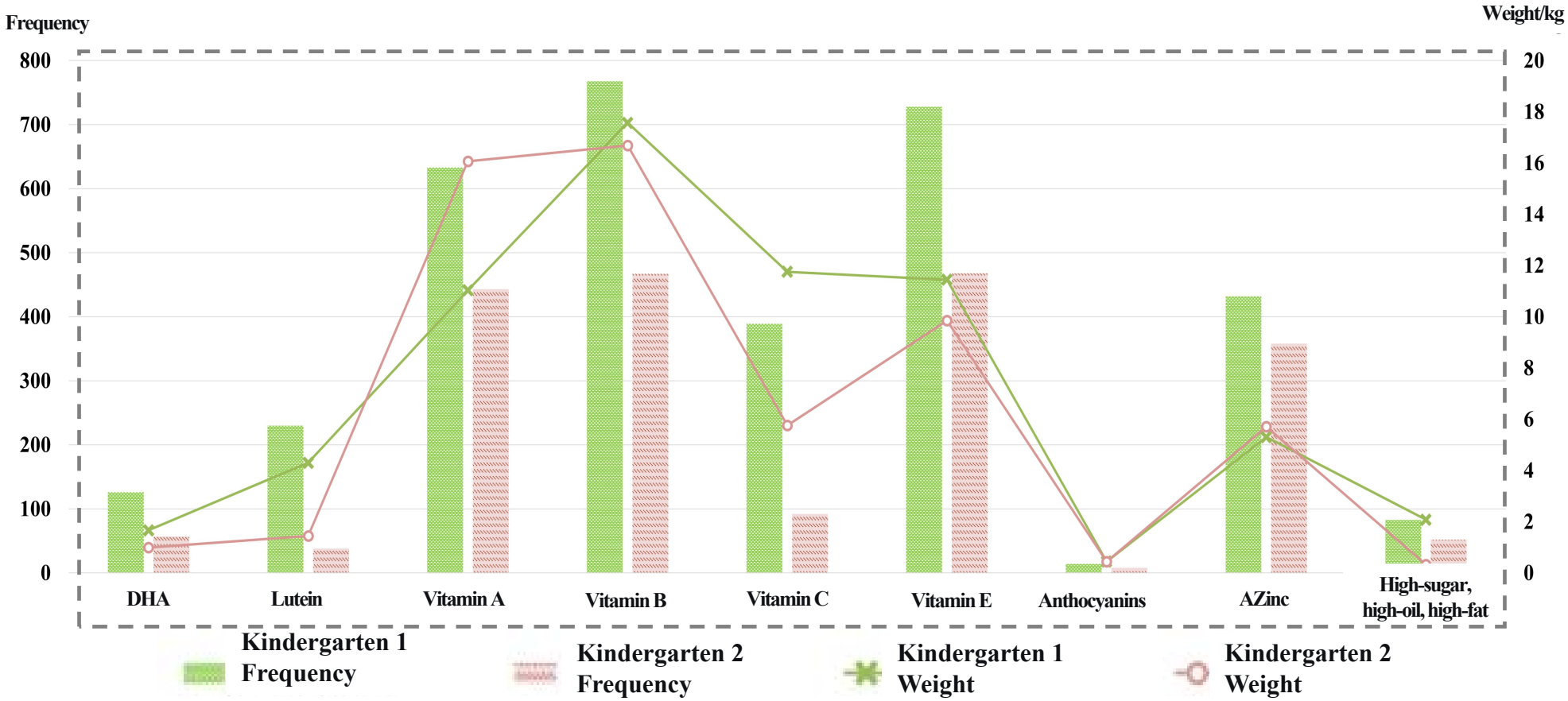
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3.2 Empirical Application

（1）Empirical Analysis of Recipes and Myopia Rates

- Is there a correlation between the recipe data and myopia rates at the two kindergartens?

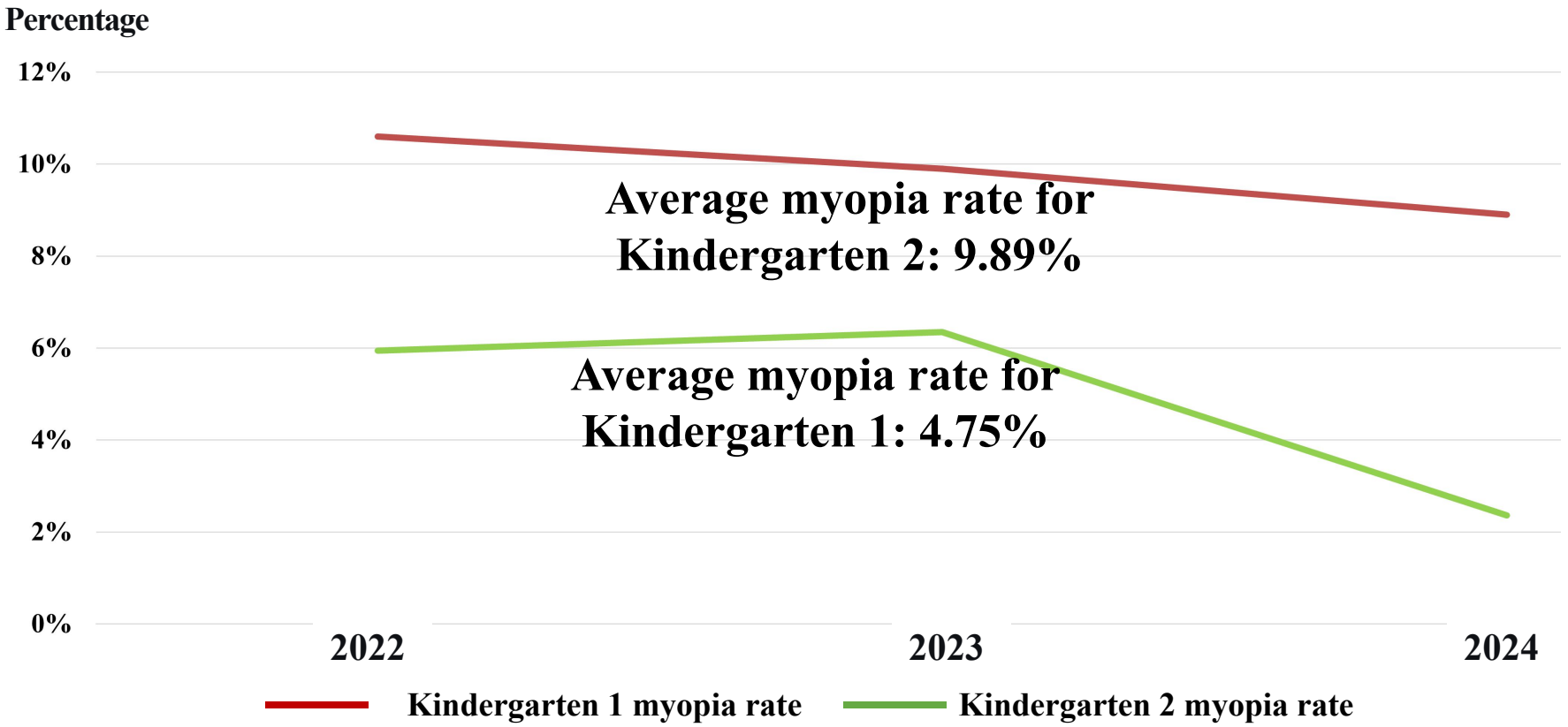


Comparison of cumulative nutrients from recipe ingredients at the two kindergartens from February to June 2024 (frequency and weight)

3.2 Empirical Application

（1）Empirical Analysis of Recipes and Myopia Rates

- Is there a correlation between the recipe data and myopia rate data from the two kindergartens?



Comparative Analysis of Student Myopia Rates at the Two Kindergartens, 2022–2024

3.2 Empirical Application

（2）Collaboration with Wuxi Sunan School Food Ingredient Company

- Presented our research findings to Sunan Company.
- Received feedback on recipe improvements from their nutrition experts.
- Sunan Company plans to incorporate myopia-preventing food-medicine homology recipes into their menus.
- Pledged long-term support for the team's practical initiatives.



Tour of Sunan Company



Presentation at Sunan Company



Sunan Company is responsible for delivering meals to 775,800 teachers and students across 719 public kindergartens and compulsory education schools in Wuxi.

3.3 Recipe Promotion

（1）1,170 Questionnaires Collected Across Three Surveys

- Survey 1: 807 responses gathered online and offline prior to the recipe design.
- Survey 2: 314 responses collected on-site during the food-medicine homology recipe lectures.
- Survey 3: Around 50 responses collected at the homemade beverage tasting event.



Surveying parents at the school gate



On-site survey after a recipe promotion lecture



On-site survey at the homemade beverage tasting event

3.3 Recipe Promotion

（2）Held Lectures at 4 Schools

- Promoted food-medicine homology recipes (streamlined from 108 down to 83 dishes).
- Introduced the "55 Questions on Dietary Therapy Knowledge for Myopia Prevention."



Lecture at Wuxi Erquan Middle School



Lecture at Wuxi Aier Experimental Kindergarten



Lecture at the Junior Division of Tianyi High School



Lecture at Wuxi Dongting Experimental Primary School

3.3 Recipe Promotion

（3）Produced 5 Bulletin Boards and 5 Roll-Up Banners

- Promoted myopia-preventing food-medicine homology recipes.
- Publicized the "55 Questions on Dietary Therapy Knowledge for Myopia Prevention."











3.3 Recipe Promotion

（4） Promotion via Douyin, Xiaohongshu, and the Sunan Company WeChat Official Account

- Posted 7 short videos created by the team on Douyin and Xiaohongshu.
- Published a popular science article on "Honey Black Sesame Milk" on the Sunan Company's WeChat Official Account, which attracted 858 views.









Wuxi Sunan Food Ingredient Company

3.3 Recipe Promotion

（5）Tasting Event Sold Out in a Flash

- Created 2 homemade beverages: "Honey Black Sesame Milk" and "Goji Berry, Chenpi, and Red Date Honey Drink."
- 100 cups of the myopia-preventing food-medicine homology beverages were snatched up in no time.
- Conducted a comparative nutritional analysis between the homemade beverages and well-known brands of milk tea and cola.



Making homemade food-medicine homology beverages



Homemade beverage tasting event



The most popular "Honey Black Sesame Milk"

3.3 Recipe Promotion

（6）Designed a Custom Carrot-Themed Team Logo

- Feedback from kindergartens and Sunan Company revealed that many children dislike eating carrots.
- We designed a carrot-themed team logo to help kids develop a liking for carrots.
- Our vision: we hope that in the future, students in Jiangsu will have eyes as sharp as an eagle's!





EaglePlate Team Logo



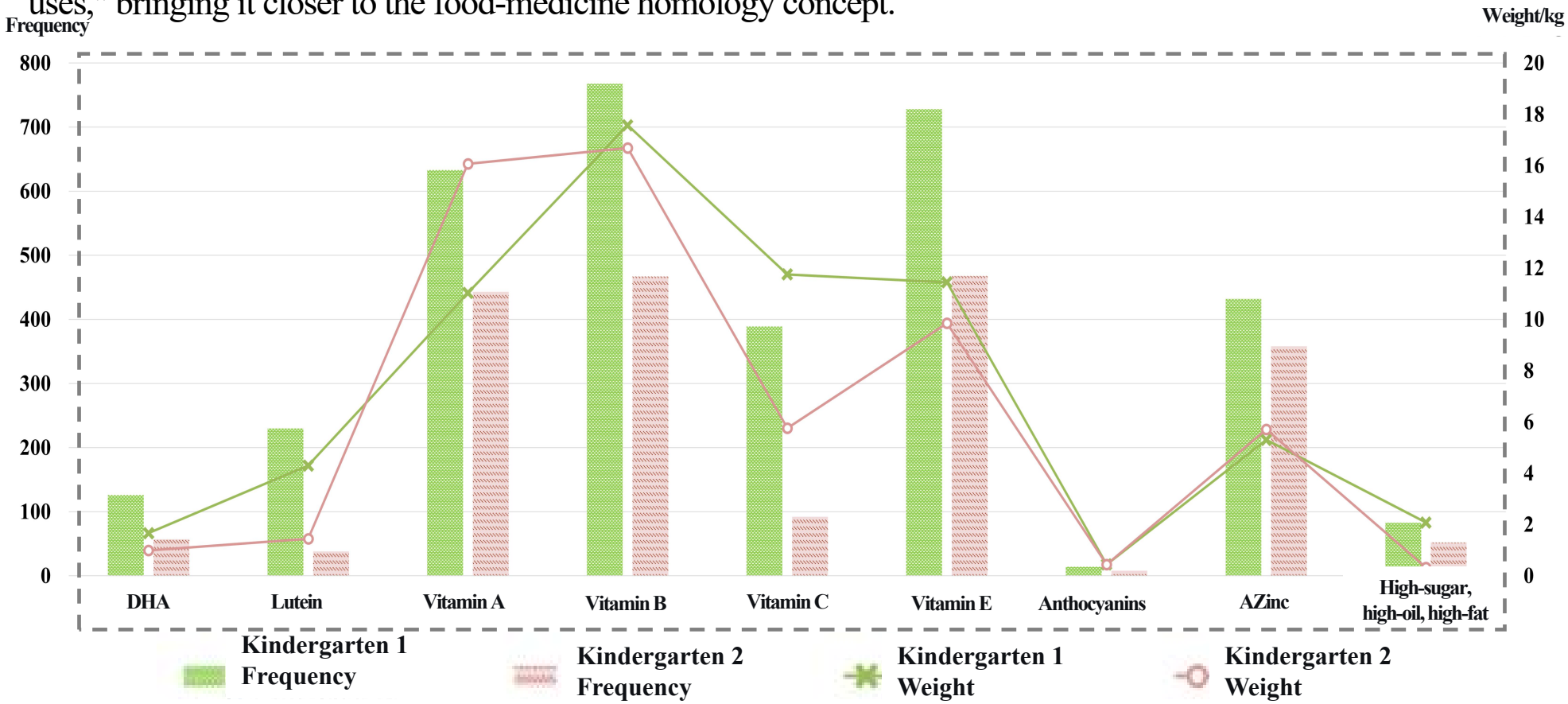
4 Results Analysis



4.1 Exploring the Correlation Between Ingredient Diversity and Myopia Rates

（1）Analysis of Kindergarten Menus from February to June 2024

- Kindergarten 1's menu featured more ingredients rich in lutein and utilized multi-functional ingredients like pumpkin and blueberries.
- Kindergarten 2's menu had a limited variety of ingredients.
- Kindergarten 1's diverse ingredients align better with the traditional Chinese medicine (TCM) philosophy of "one ingredient, multiple uses," bringing it closer to the food-medicine homology concept.

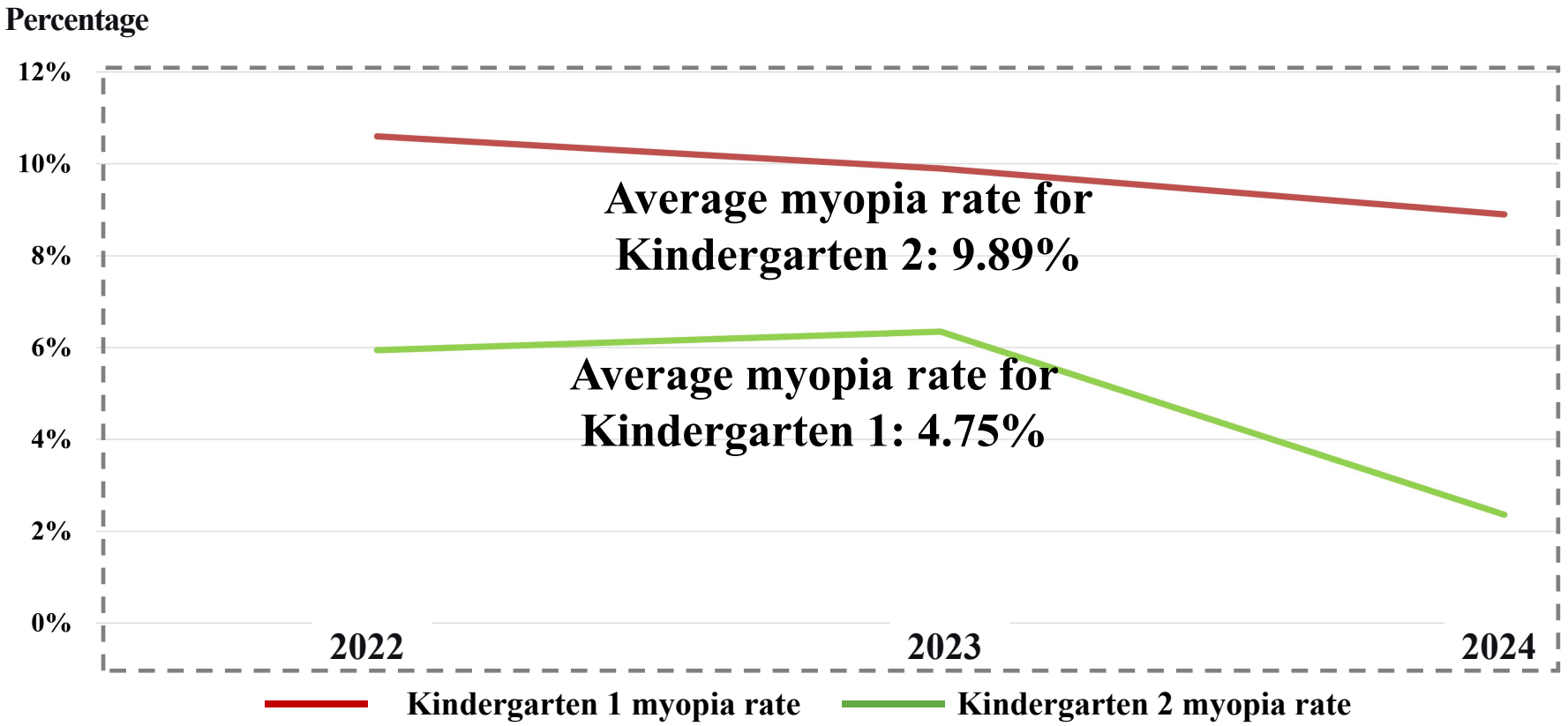


Comparison of cumulative nutrients from recipe ingredients at the two kindergartens from February to June 2024 (frequency and weight)

4.1 Exploring the Correlation Between Ingredient Diversity and Myopia Rates

（2）Analysis of Student Myopia Rates at the Two Kindergartens (2022–2024)

- Kindergarten 1's average myopia rate (4.75%) was significantly lower than Kindergarten 2's (9.89%).
- In 2024, Kindergarten 1 showed a notably steeper decline in myopia rate compared to Kindergarten 2.



Comparative Analysis of Student Myopia Rates at the Two Kindergartens, 2022–2024

4.1 Exploring the Correlation Between Ingredient Diversity and Myopia Rates

（3）Conclusions from the Empirical Analysis

- The more diverse the nutritional profile of ingredients in a menu, the lower the myopia rate.
- School cafeterias in Jiangsu lack myopia-preventing food-medicine homology recipes.
- Existing cafeteria menus need improvement.

According to the self-compiled "55 Questions on Dietary Therapy for Myopia Prevention," what are the shortcomings of current menus?

表 2 无锡苏南学校食材配送公司现有食谱：食谱 1

菜餐1	日期	星期一			星期二			星期三			星期四			星期五		
	类别	品名	菜名	克重(g)	品名	菜名	克重(g)	品名	菜名	克重(g)	品名	菜名	克重(g)	品名	菜名	克重(g)
	学生副餐	纯牛奶	纯牛奶125ml	1	奶香餐包	奶香餐包40g/个	1	纯牛奶	纯牛奶125ml	1	南瓜籽仁蛋糕	南瓜籽仁蛋糕 40g/个	1	香蕉	进口香蕉（个）	1
	主食	软米饭	粳米	120	玉米糁饭	玉米糁150g/包	3	软米饭	粳米	120	燕麦饭	燕麦片150g/包	3	软米饭	粳米	120
						粳米	117					粳米	117			
	大荤	红烧鸭边腿块	冰鲜鸭边腿块	150	清炖/红烧骨块	冷鲜扇骨块	145	咖喱牛肉块	国产冷鲜牛肉块	100	时蔬炒虾仁	大圆椒块	30	肉饼蒸蛋	冷鲜猪肉末	70
									新土豆块	30		胡萝卜块	20		鸡蛋（清洁蛋）	40
									胡萝卜块	15		冰鲜虾仁	50			
												鸡蛋（清洁蛋）	5			
												洋葱丝	15			
	小荤	花菜胡萝卜肉片	松花菜朵	60	西红柿炒蛋	西红柿	90	糖醋里脊肉条	猪肉	45	梅汁翅根	冰鲜鸡翅根（50-60g/个）	120	荷苗炒肉片	荷苗片	70
			冷鲜猪精肉片	40		鸡蛋（清洁蛋）	45								冷鲜猪精肉片	40
			胡萝卜片	10												
	素菜	菠菜百叶	菠菜段	85	葱油卜	白萝卜块	105	娃娃菜姬菇	娃娃菜段	90	毛白菜油三角	毛白菜段	100	红烧粉丝	冷鲜猪肉末	5
			百叶	5					姬菇	5		油三角	5		红薯粉丝500g/包	18
	汤	鸡蛋酸不辣汤	鸡蛋（清洁蛋）	15	白菜肉丝汤	白菜段	25	肉末香菇豆腐汤	嫩豆腐	20	西湖牛肉羹	国产冷鲜牛肉末	5	龙骨萝卜汤	冷鲜带肉龙骨	30
			白菜丝	15		冷鲜猪肉丝	10		去根香菇	8		金针菇	10		白萝卜块	40
			红薯粉丝500g/包	3					冷鲜猪肉末	10		鸡蛋（清洁蛋）	10		大蒜	2
			去根香菇	10								香菜（净）	1			
	配料	配料	细米葱	1	配料	细米葱	2	配料	细米葱	1	配料	细米葱	1.5	配料	细米葱	1
			老姜	1.5		老姜	1.5		老姜	1.5		老姜	1.5		老姜	1.5
			大蒜籽	1		大蒜籽	1		大蒜籽	1		大蒜籽	1.5		大蒜籽	1

Current menus for primary and secondary schools in Wuxi provided by Sunan Company

4.2 Identifying Directions for Menu Improvement

（1）Five Shortcomings of the Current Menus

- Dishes are overly rich and heavy in flavor.
- They lack key nutrients such as lutein.
- They lack ingredients rich in DHA.
- The sources of lutein are limited.
- They lack hard-textured foods.

Improving existing menus with reference to our self-compiled "83 Recipes"

表 2 无锡苏南学校食材配送公司现有食谱：食谱 1

套餐1	日期	星期一			星期二			星期三			星期四			星期五		
	类别	品名	菜名	克重（g）	品名	菜名	克重（g）	品名	菜名	克重（g）	品名	菜名	克重（g）	品名	菜名	克重（g）
	学生副餐	纯牛奶	纯牛奶125ml	1	奶香餐包	奶香餐包40g/个	1	纯牛奶	纯牛奶125ml	1	南瓜籽仁蛋糕	南瓜籽仁蛋糕40g/个	1	香蕉	进口香蕉（个）	1
	主食	软米饭	粳米	120	玉米糁饭	玉米糁150g/包	3	软米饭	粳米	120	燕麦饭	燕麦片150g/包	3	软米饭	粳米	120
						粳米	117					粳米	117			
	大荤	红烧鸭边腿块	冰鲜鸭边腿块	150	清炖/红烧骨块	冷鲜扇骨块	145	咖喱牛肉块	国产鲜牛肉块	100	时蔬炒虾仁	大圆椒块	30	肉饼蒸蛋	冷鲜猪肉末	70
									新土豆块	30		胡萝卜块	20		鸡蛋（清洁蛋）	40
									胡萝卜块	15		冰鲜虾仁	50			
												鸡蛋（清洁蛋）	5			
												洋葱丝	15			
	小荤	花菜胡萝卜肉片	松花菜朵	60	西红柿炒蛋	西红柿	90	糖醋里脊肉条	猪肉	45	梅汁翅根	冰鲜鸡翅根（50-60g/个）	120	黄豆炒肉片	黄豆片	70
			冷鲜猪肉片	40		鸡蛋（清洁蛋）	45								冷鲜猪精肉片	40
			胡萝卜片	10												
	素菜	腰菜百叶	腰菜段	85	葱油萝卜	白萝卜块	105	娃娃菜烩蘑菇	娃娃菜段	90	白油三角	毛白菜段	100	红烧粉丝	冷鲜猪肉末	5
			百叶	5					蘑菇	5		油三角	5		红薯粉丝500g/包	18
	汤	鸡蛋酸不辣汤	鸡蛋（清洁蛋）	15	白菜肉丝汤	白菜段	25	肉末香菇豆腐汤	嫩豆腐	20	西湖肉羹	国产冷鲜牛肉末	5	龙骨萝卜汤	冷鲜带肉龙骨	30
			白菜丝	15		冷鲜猪精肉丝	10		去根香菇	8		金针菇	10		白萝卜块	40
			红薯粉丝500g/包	3					冷鲜猪肉末	10		鸡蛋（清洁蛋）	10		大蒜	2
			去根香菇	10								香菜（净）	1			
	配料	配料	细米葱	1	配料	细米葱	2	配料	细米葱	1	配料	细米葱	1.5	配料	细米葱	1
			老姜	1.5		老姜	1.5		老姜	1.5		老姜	1.5		老姜	1.5
			大蒜籽	1		大蒜籽	1		大蒜籽	1		大蒜籽	1.5		大蒜籽	1

Current menus for primary and secondary schools in Wuxi provided by Sunan Company

4.2 Identifying Directions for Menu Improvement

（2） Five Features of Myopia-Preventing Food-Medicine Homology Recipes

- Include more eye-nourishing, food-medicine homology ingredients.
- Draw on the Jiangnan dietary pattern.
- Adopt healthy cooking methods such as steaming and stewing.
- Increase ingredient diversity.
- Incorporate hard-textured foods.

Improved the current menus with reference to the self-compiled "83 Recipes"

表 3 无锡苏南学校食材配送有限公司改进食谱：食谱 2

类别	星期一	星期二	星期三	星期四	星期五
上午点心	纯牛奶 食部：纯牛奶：125ml 黑芝麻花卷 食部：芝麻子：1.5g 食部：小麦粉：10g	纯牛奶 食部：纯牛奶：125ml 薄脆饼干 食部：饼干：12g	纯牛奶 食部：纯牛奶：125ml 蒸紫薯 食部：紫薯：60g	纯牛奶 食部：纯牛奶：125ml 苏打饼干 食部：饼干：12g	纯牛奶 食部：纯牛奶：125ml 蒸玉米 市品：玉米：60g
午餐	食部：精盐：1g 食部：油：6.5g 燕麦饭 食部：燕麦：10g 食部：粳米：45g 红烧鸡翅 市品：鸡翅：70g 塌菜油三角 食部：塌菜：75g 食部：油豆腐：5g 菠菜猪肝汤 食部：菠菜：15g 食部：猪肝：15g 食部：胡萝卜：5g	食部：精盐：1 食部：油：6.5 软米饭 食部：粳米：55 清蒸鲳鱼 市品：鲳：50g 西芹炒香干 食部：豆腐干：25g 食部：西芹：40g 西红柿蛋汤 食部：奶柿子：30g 市品：鸡蛋：10g	食部：精盐：1g 食部：油：6.5g 胡萝卜米饭 食部：粳米：45g 食部：胡萝卜：10g 糖醋里脊肉条 市品：猪肉：45g 时蔬炒双花 食部：白花菜：30g 食部：甜椒：15g 食部：西兰花：30g 鸡毛菜腐皮蛋汤 市品：鸡蛋：10g 食部：豆腐皮：1g 食部：鸡毛菜：12g	食部：精盐：1g 食部：油：6.5g 软米饭 食部：粳米：55g 土豆炖牛肉 食部：马铃薯：10 市品：牛肉：45g 食部：胡萝卜：20g 素炒黄瓜 食部：黄瓜：75g 芳苣木耳蛋汤 食部：干黑木耳：0.5g 食部：芳笋：25g 市品：鸡蛋：10g	食部：精盐：1g 食部：油：6.5g 番茄肉酱意大利面 食部：面条：40g 食部：洋葱：15g 食部：奶柿子：40g 市品：猪肉：20g 萝卜排骨汤 市品：猪小排：20g 食部：白萝卜：30g
下午点心	牛肉粉丝汤 食部：平菇：10g 食部：小白菜：10g 食部：粉丝：12g 食部：牛肉：9g 耙耙柑 市品：丑橘：100g	枸杞南瓜汤 市品：梨：90g 市品：银耳：0.2g 食部：枸杞子：1.5g 市品：枣：5g 每日坚果 食部：每日坚果：12.5g 贝贝南瓜 市品：南瓜：60g	三鲜丁丁面 市品：猪肉：9g 食部：面条：25g 食部：奶柿子：20g 食部：生菜：10g 千禧 市品：橄榄番茄：80g	小米山药枸杞粥 食部：小米：15g 食部：枸杞子：1.5g 市品：山药：25g 蓝莓 食部：蓝莓：40g	燕斯风味酸奶 食部：酸奶：125ml 手工鸡蛋糕 食部：鸡蛋糕：40g 苹果 市品：苹果：90g

The improved food-medicine homology menus by Sunan Company

4.3 Analyzing the Nutritional Composition of the Homemade Drinks at the Tasting Event

（1）Nutritional Analysis of the Two Homemade Drinks

- The homemade drinks are rich in vision-beneficial nutrients such as calcium, iron, zinc, and vitamins A, B1, and B2.
- They contain no artificial additives.
- The sugars are mostly naturally occurring fructose, which is easily absorbed.

Nutritional analysis conducted under the guidance of a nutritionist from Sunan Company

Preparation Ingredients:		
Ingredient	Weight	Unit
Pure milk	1000	ml
Roasted black sesame	50	g
Honey	50	g
Preparation Method:		
1. First, add 1000 ml of milk to a blender or soy milk maker; 2. Then add 50 g ground roasted black sesame and 50 g honey; 3. Use the soy milk mode to blend and heat; once cooked, it is ready to drink.		
Note: Food–medicine-homology ingredients include black sesame and honey.		
Nutrient Analysis:		
Nutrient	Intake	Unit
Protein (g)	44.75	g
Energy (kJ)	3809	kJ
Calcium (mg)	1502	mg
Iron (mg)	13.85	mg
(mg)	5.25	mg
Phosphorus (mg)	1319.5	mg
Vitamin A (μg)	450	μg
Thiamine B1 (mg)	0.53	mg
Riboflavin B2 (mg)	1.75	mg
Niacin (mg)	3.7	mg
Ascorbic acid/C (mg)	1.5	mg
Main Nutrient Ratio Relationships:		
Nutrient	Requirement %	Actual %
Protein	8–20	19.43%
Fat	20–30	38.10%
Animal fat	30–50	38.46%
Carbohydrate	50–65	42.47%

Nutritional analysis of Honey Black Sesame Milk

Preparation Ingredients:		
Ingredient	Weight	Unit
Goji berries	40	g
Jujubes	100	g
Chenpi (dried tangerine peel)	20	g
Honey	50	g
Purified water	2000	ml
Preparation Method:		
1. First, add 1000 ml of purified water to a pot, then add 40 g of goji berries, 100 g of jujubes (washed and cut open), 20 g of chenpi (washed), and 50 g of honey in sequence; 2. After bringing to a boil over high heat, simmer over low heat for 20 minutes; strain the decoction and set aside; 3. Add another 1000 ml of purified water to the pot, bring to a boil, then simmer for 20 minutes; strain and pour into the first batch of decoction, stir well, and it is ready to drink.		
Note: Food–medicine-homology ingredients include goji berries, jujubes, chenpi, and honey.		
Nutrient Analysis:		
Nutrient	Intake	Unit
Protein (g)	9.46	g
Energy (kJ)	2663	kJ
Calcium (mg)	96.4	mg
Iron (mg)	6.4	mg
Zinc (mg)	1.427	mg
Phosphorus (mg)	136.1	mg
Vitamin A (μg)	720	μg
Thiamine B1 (mg)	0.22	mg
Riboflavin B2 (mg)	0.359	mg
Niacin (mg)	3.65	mg

Nutritional analysis of Goji Berry, Chenpi, and Red Date Honey Drink

4.3 Analyzing the Nutritional Composition of the Homemade Drinks at the Tasting Event

（2）Nutritional Analysis of Well-Known Brands of Milk Tea and Cola

- Beverages like milk tea and cola lack labeling for key nutrients.
- They contain artificial additives.
- As high-sugar beverages with added sweeteners, they pose a myopia risk.

Conclusion: "Honey Black Sesame Milk" and "Goji Berry, Chenpi, and Red Date Honey Drink" are more conducive to lowering the risk of myopia!

Table 3 Nutritional Composition Analysis of a Well-Known Brand of Taro-Flavored Milk Tea

Ingredients:

Milk tea powder ingredients: non-dairy creamer (glucose syrup, refined vegetable oil, sodium caseinate, mono- and diglycerides of fatty acids, dipotassium hydrogen phosphate, silicon dioxide), white granulated sugar, edible glucose, skimmed milk powder, instant black tea powder, instant oolong tea powder, whey protein powder, food additives (tricalcium phosphate), edible flavors and fragrances.

Milk tea fruit ingredients: water, coconut fiber, high-fructose corn syrup, white granulated sugar, edible glucose, food additives (citric acid, sodium citrate, sodium carboxymethyl cellulose, guar gum, calcium lactate, potassium sorbate).

Nutrient Analysis:

Nutrient	Intake (per 80 g total)	Unit
Energy	901	kJ
Protein	1.2	g
Fat	7.5	g
Carbohydrate	35.5	g
Sodium	67	mg

Table 4 Nutritional Composition Analysis of a Well-Known Brand of Cola

Ingredients:

Main ingredients: water, high-fructose corn syrup, white granulated sugar, food additives (carbon dioxide, caramel color, phosphoric acid, caffeine), edible flavors, and fragrances.

Nutrient Analysis:

Nutrient	Intake (per 100 ml)	Unit
Energy	180	kJ
Protein	0	g
Fat	0	g
Carbohydrate	10.6	g
Sugars	10.6	g
Sodium	12	mg

Nutritional analysis of a well-known brand of milk tea and a well-known brand of cola



5 Discussion and Conclusion





5 Discussion and Conclusion

- Compiled 83 myopia-preventing recipes and summarized 55 key points of dietary therapy knowledge for myopia.
- Empirical analysis revealed a correlation between recipe ingredient diversity and myopia rates.
- Survey analysis showed that school cafeterias lack myopia-preventing food-medicine homology recipes.
- Identified 5 key features of myopia-preventing food-medicine homology recipes.
- Launched a multi-channel promotion campaign: "Lectures + Posters + Short Videos + Popular Science Articles + Tastings."
- 95% of students approved of the myopia-preventing food-medicine homology beverages.

Exploring the science of myopia prevention, promoting traditional Chinese medicine dietary therapy culture, and driving innovation in campus catering.

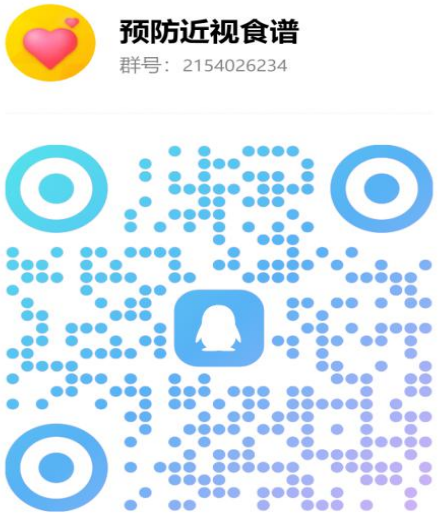
We've been called "idiots" to our faces, received plenty of cold rejections, struggled to understand academic papers, failed to get data, didn't know how to use the software, waded through tedious organization, and even argued with each other...
But what matters is that we kept going together for 141 days.



October 12, 2024: EaglePlate Team's First Meeting

EaglePlate Team Slogan

Promoting food-medicine homology recipes to protect a bright, healthy future for students' eyes!



Scan the QR code to join our [QQ Group]:

2154026234